

Oscillators

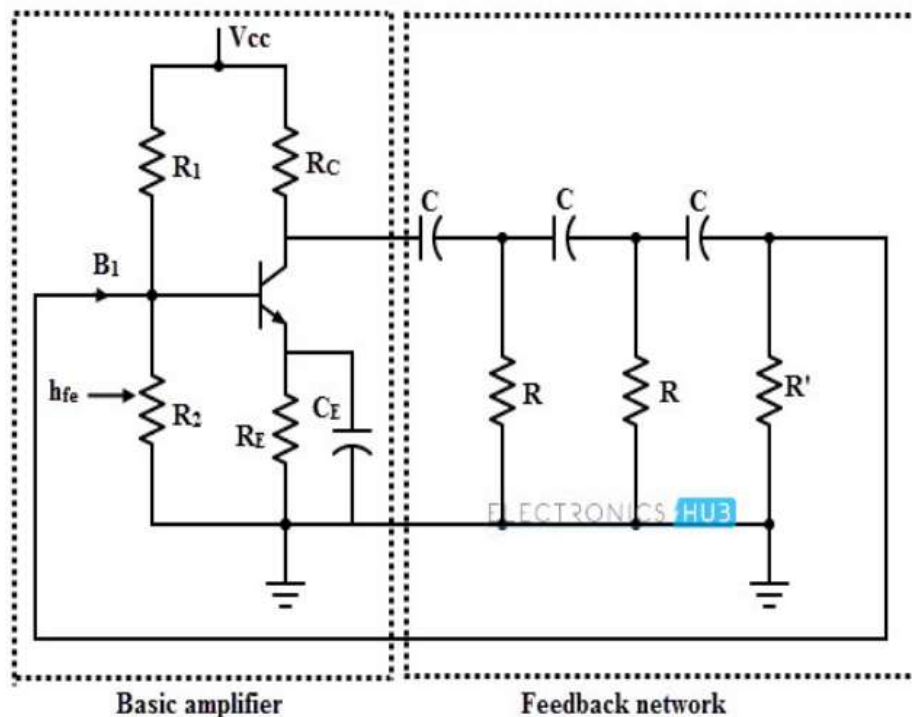
An **RC Phase Shift Oscillator** is a type of electronic oscillator that uses **resistors (R)** and **capacitors (C)** in a feedback network to produce a **sinusoidal waveform**. It is commonly used to generate low-frequency signals (up to a few hundred kHz) and is notable for its **simplicity and ease of design**.

🔧 Basic Working Principle

The RC phase shift oscillator consists of:

1. **An amplifier** (usually an op-amp or transistor) — provides the necessary gain.
2. **A feedback network** — usually three cascaded RC sections connected in series.

Each RC stage provides a phase shift, and three of them together can provide a total of **180° phase shift**. The amplifier also inverts the signal (another **180° shift**), making the **total phase shift 360° (or 0°)**, satisfying the **Barkhausen criterion** for sustained oscillations.



□ Theory and Design Considerations

Barkhausen Criterion

- **Loop gain ≥ 1**
- **Total phase shift around the loop = 360° or 0°**

Frequency of Oscillation

For a 3-stage RC network:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

Where:

- RR and CC are the resistance and capacitance in each stage.

Gain Requirement

The amplifier must compensate for attenuation in the RC network. In a transistor circuit, the gain AA should be **greater than or equal to 29**.

Applications

- Audio tone generation
 - Signal generators
 - Function generators
 - Low-frequency oscillator circuits
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Advantages

- Simple design
 - Requires no inductors (suitable for IC implementation)
 - Stable frequency with proper component selection
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Limitations

- Not suitable for high-frequency oscillations
 - Gain needs to be carefully controlled for distortion-free output
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□ Overview: Wien Bridge Oscillator Using Transistors

A **Wien Bridge Oscillator with transistors** is an analog oscillator circuit that uses **two transistors** (usually in a **two-stage amplifier** configuration) to provide the required gain and phase conditions. The **Wien bridge network** (made of resistors and capacitors) provides **positive feedback**, while another path controls **negative feedback** to stabilize amplitude.

▴ Basic Circuit Components

1. **Two Transistors (Q1 & Q2):**
 - Form a two-stage amplifier (commonly in **common emitter** configuration).
 - Provide overall **180° phase shift**.
 2. **RC Wien Bridge Network** (positive feedback):
 - A **series RC** (R1–C1) branch and a **parallel RC** (R2–C2) branch.
 - Provides **0° phase shift** at resonance.
 3. **Negative Feedback Path:**
 - Gain control via resistors or sometimes nonlinear elements like thermistors or bulbs.
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⚙ How It Works

- The **first transistor amplifies** the input.
 - The **second transistor further amplifies** and provides necessary phase shift.
 - The **RC network selects the frequency** where the phase shift is 0°, and the amplitude is fed back **in phase** to sustain oscillations.
 - The **feedback must meet Barkhausen Criterion:**
 - Total loop gain = 1
 - Total phase shift = 360° (or 0°)
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🔋 Biasing and Stability

- Both transistors are **properly biased** using resistors to operate in the active region.
- The **gain is carefully adjusted** to just start and sustain oscillations.
- For stable output without distortion, some designs include **automatic gain control (AGC)** using thermistors or diodes.

✓ Advantages

- Good for analog signal generation without op-amps
- Can be implemented in discrete component designs

✗ Disadvantages

- More components and careful biasing than op-amp version
- Temperature stability can be an issue without AGC

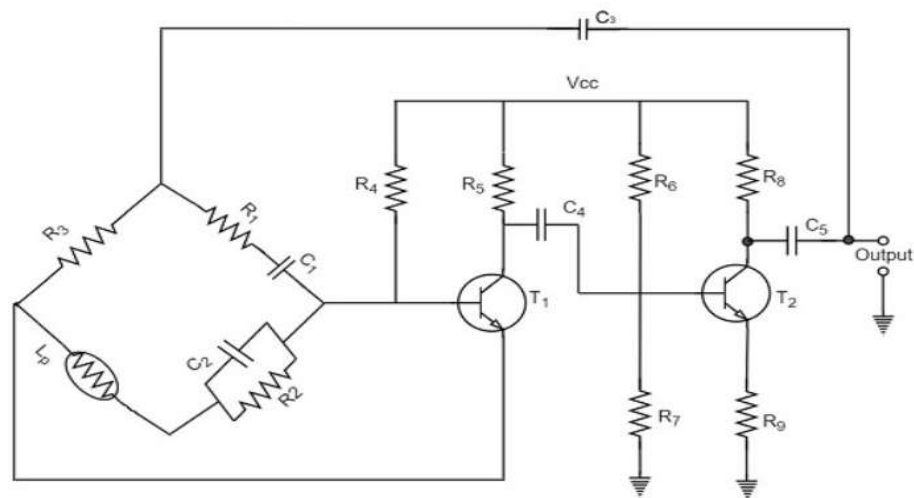
□ Typical Application

- Lab signal generators
- Audio tone generation
- Reference sine wave source in analog circuits

Construction

The circuit construction of Wien bridge oscillator can be explained as below. It is a two-stage amplifier with RC bridge circuit. The bridge circuit has the arms R_1C_1 , R_3 , R_2C_2 and the tungsten lamp L_p . Resistance R_3 and the lamp L_p are used to stabilize the amplitude of the output.

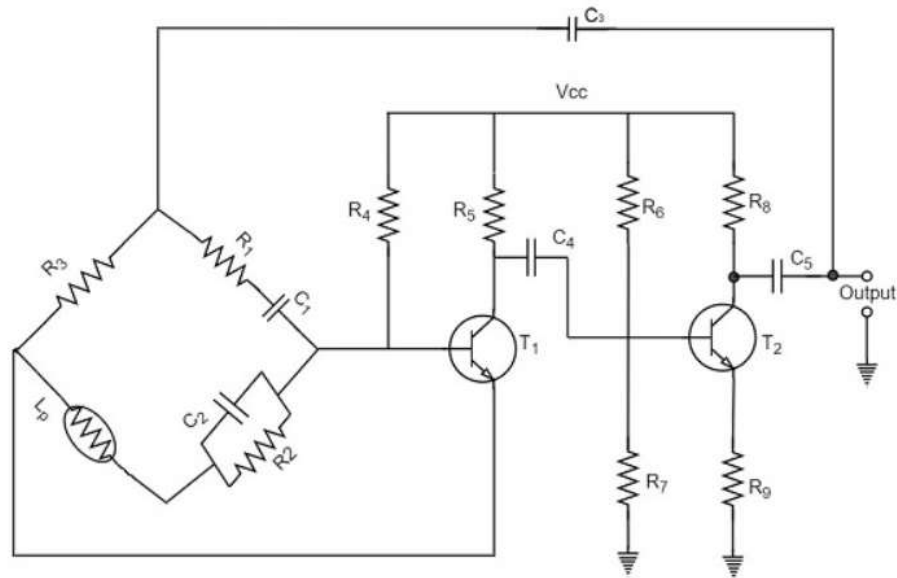
The following circuit diagram shows the arrangement of a Wien bridge oscillator.



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The following circuit diagram shows the arrangement of a Wien bridge oscillator.



The transistor T_1 serves as an oscillator and an amplifier while the other transistor T_2 serves as an inverter. The inverter operation provides a phase shift of 180° . This circuit provides positive feedback through R_1C_1 , C_2R_2 to the transistor T_1 and negative feedback through the voltage divider to the input of transistor T_2 .

The frequency of oscillations is determined by the series element R_1C_1 and parallel element R_2C_2 of the bridge.

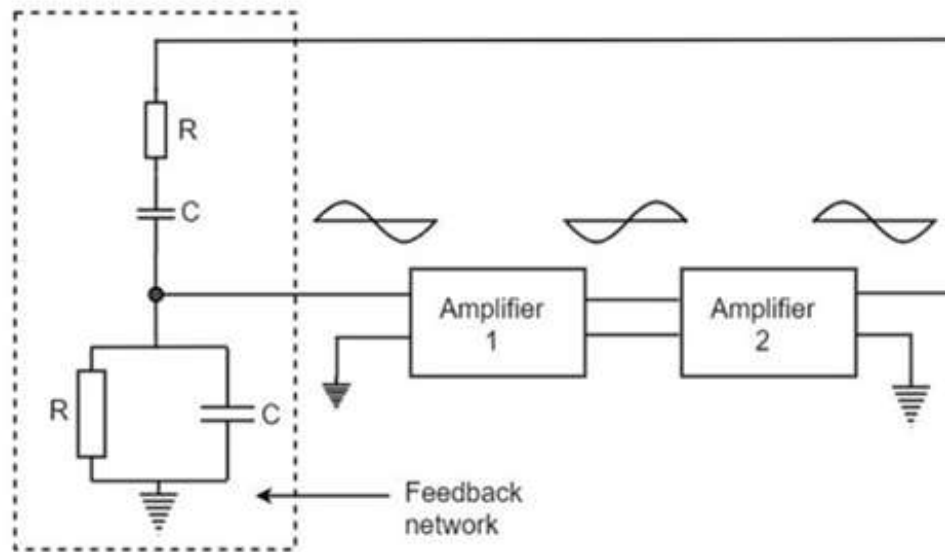
$$f = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}}$$

If $R_1 = R_2$ and $C_1 = C_2 = C$

Then,

$$f = \frac{1}{2\pi RC}$$

Now, we can simplify the above circuit as follows –



The oscillator consists of two stages of RC coupled amplifier and a feedback network. The voltage across the parallel combination of R and C is fed to the input of amplifier 1. The net phase shift through the two amplifiers is zero.

The usual idea of connecting the output of amplifier 2 to amplifier 1 to provide signal regeneration for oscillator is not applicable here as the amplifier 1 will amplify signals over a wide range of frequencies and hence direct coupling would result in poor frequency stability. By adding Wien bridge feedback network, the oscillator becomes sensitive to a particular frequency and hence frequency stability is achieved.

crystal oscillator

A **crystal oscillator** is an electronic oscillator circuit that uses the mechanical resonance of a vibrating **piezoelectric crystal**, typically **quartz**, to create an electrical signal with a **precise frequency**. This frequency is commonly used to keep track of time (in watches and clocks), to provide a stable clock signal for digital integrated circuits, and to stabilize frequencies for radio transmitters and receivers.

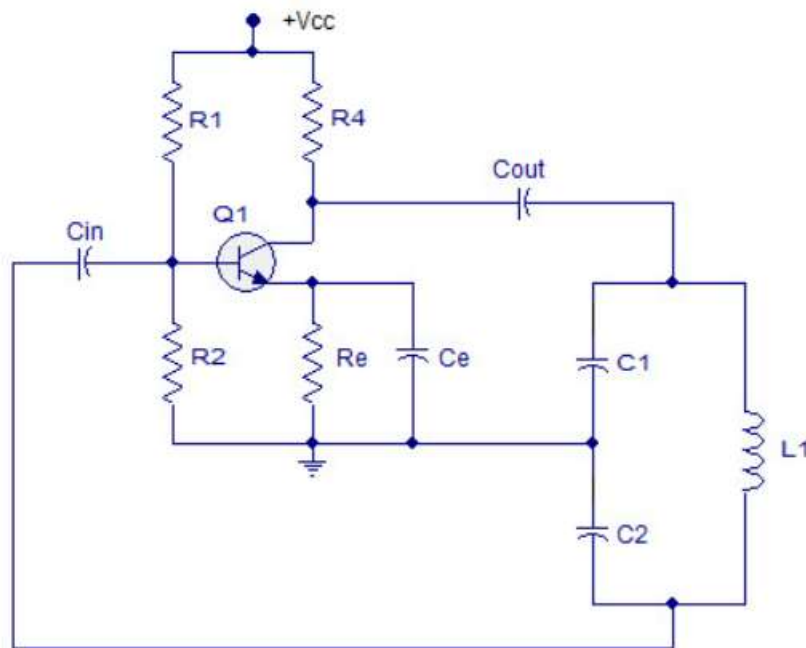
□ Basic Principle

The crystal oscillator operates based on the **piezoelectric effect**:

- When a **voltage** is applied to a quartz crystal, it **vibrates**.
 - Conversely, when the quartz crystal vibrates, it generates an **AC voltage**.
 - Quartz is cut in a specific way (AT-cut is common) so that it resonates at a **predictable, stable frequency** when energized.
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⚙️ Working of a Crystal Oscillator

1. **Excitation Circuit:**
 - An **amplifier** (e.g., using a transistor or op-amp) applies an AC signal to the crystal.
2. **Crystal Resonance:**
 - The crystal behaves like a highly selective bandpass filter.
 - It allows only its **natural resonant frequency** to pass through and blocks others.
3. **Positive Feedback:**
 - The output is fed back to the input through the crystal to maintain oscillation.
 - This satisfies the **Barkhausen criteria** for sustained oscillations.
4. **Stable Output:**
 - The oscillating signal is available at the output terminal.
 - It is very **stable, low noise, and accurate**.



□ Types of Crystal Oscillator Circuits

1. **Pierce Oscillator** (most common in microcontrollers)
 2. **Colpitts Crystal Oscillator**
 3. **Hartley Crystal Oscillator**
 4. **Clapp Oscillator**
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Advantages

- Very **high frequency stability**
 - **Low phase noise**
 - **Temperature-compensated** and **voltage-stable**
 - Widely available and low-cost
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Applications

- **Microcontrollers and CPUs** (for clock generation)
 - **Watches and clocks**
 - **Radio transmitters/receivers**
 - **Communication systems**
 - **Signal generators**
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⚙️ □ 2. Working Principle

1. **Power is applied**, and the transistor is biased into the active region.
 2. The **quartz crystal** oscillates at its natural resonant frequency.
 3. The **feedback network (C1 and C2)** forms a capacitive voltage divider, providing **positive feedback** to the base of the transistor.
 4. If the **Barkhausen criteria** are met:
 - Total **loop gain** ≥ 1
 - Total **phase shift** = 0° or $360^\circ \rightarrow$ sustained oscillations occur.
 5. The **crystal locks the frequency**, ensuring high stability and accuracy.
 6. The output is typically taken from the **collector** of the transistor.
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✓ 3. Advantages

- **High frequency stability** (thanks to the crystal)
 - **Simple design**
 - **Low phase noise**
 - **High Q-factor** due to the crystal
 - **Reliable startup**
 - Works well in **high-frequency ranges**
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⚠ 4. Limitations

- **Fixed frequency** (set by the crystal)
 - Not suitable for **wide range tuning**
 - Limited output power
 - Sensitive to component variation if poorly designed
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✂ 5. Applications

- **Clock generation** in microcontrollers and digital ICs
- **RF oscillators** and **radio transmitters**
- **Communication systems**
- **Instrumentation** and **measurement equipment**
- **Watches and timers**
- Any application requiring **precise frequency generation**